
DEPARTMENT OF MATHEMATICS

Mathematics Lab Assessment No. 4

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Question	In automated manufacturing systems, 100% inspection of every produced component is often impractical due to time, cost, and throughput constraints. Engineers rely on samples to draw conclusions about the entire production process. However, sample-based conclusions carry inherent uncertainty — a sample mean may differ from the true population mean simply due to random variation. This experiment addresses the following engineering questions using statistical inference: • Has the process mean shifted from the target specification? • How confident are we in our estimate of the true mean? • If we conclude the process has drifted, what is the risk that we are wrong? • How many samples should we collect to achieve a desired level of precision? Students will work with three distinct datasets representing different automation scenarios, applying progressively more advanced inference techniques to each. Dataset 1: Robotic Welding Station — Cycle Time Validation A robotic welding station has a target cycle time of 12.0 seconds. After recalibration, 20 randomly selected cycles are timed (in seconds): 12.3, 11.8, 12.5, 12.1, 11.9, 12.4, 12.0, 12.2, 11.7, 12.6, 12.1, 12.3, 11.8, 12.0, 12.5, 12.2, 11.9, 12.4, 12.1, 12.3 Target (μ_0) = 12.0 sec. Population σ is unknown. Dataset 2: Automated Filling System — Volume Compliance An automated liquid filling machine is designed to dispense exactly 500 mL per bottle. The quality team suspects overfilling (which wastes material). A sample of 25 bottles is measured (in mL): 501.2, 500.8, 502.1, 499.5, 501.5, 500.3, 501.8, 500.0, 502.3, 499.7, 501.0, 500.6, 501.9, 500.2, 501.4, 500.9, 501.7, 500.1, 501.3, 500.5, 501.6, 500.4, 501.1, 500.7, 501.2 Target (μ_0) = 500.0 mL. Population σ is unknown. Use a one-sided (right-tailed) test.
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	Dataset 3: CNC Precision Boring — Diameter Tolerance A CNC boring machine produces holes with a specified diameter of 25.00 mm. Historical data establishes $\sigma = 0.05$ mm (known). A sample of 30 holes is measured (in mm): 25.02, 24.98, 25.01, 25.03, 24.99, 25.00, 25.04, 24.97, 25.01, 25.02, 25.00, 24.99, 25.03, 25.01, 24.98, 25.02, 25.00, 24.97, 25.01, 25.03, 25.02, 24.99, 25.00, 25.01, 24.98, 25.03, 25.02, 25.00, 24.99, 25.01 Target (μ_0) = 25.00 mm. Population $\sigma = 0.05$ mm (known). Use a two-sided Z-test.
Solution	i) Identify the parameters and mathematical concept Dataset 1 (Robotic Welding Station) - Target Mean (μ_0) = 12.0 s - Sample Size (n) = 20 - Sample Mean ($\bar{x}$) - Sample Standard Deviation (s) Dataset 2 (Automated Filling System) - Target Mean (μ_0) = 500 mL - Sample Size (n) = 25 - Sample Mean ($\bar{x}$) - Sample Standard Deviation (s) Dataset 3 (CNC Boring Machine) - Target Mean (μ_0) = 25.00 mm - Population Standard Deviation (σ) = 0.05 mm - Sample Size (n) = 30 - Sample Mean ($\bar{x}$) Mathematical Models Confidence Interval Estimation - t-Interval (σ unknown) - Z-Interval (σ known) Hypothesis Testing - Two-Sided t-Test (Dataset 1) - One-Sided t-Test (Dataset 2) - Two-Sided Z-Test (Dataset 3) Error Analysis - Type I Error (α) - Type II Error (β) Sample Size Determination $n = (Z_{\alpha/2} \sigma / E)^2$

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ii) Solve analytically

Dataset 1:

$$n=20, \mu_0 = 125, \alpha = 0.05$$

$$\bar{x} = 12.1550$$

$$s = 0.2585$$

$$SE = \frac{s}{\sqrt{n}} = \frac{0.2585}{\sqrt{20}} = 0.0578$$

$$95\% \quad t_{0.025, 19} = 2.093$$

$$ME = t \times SE = 2.093(0.0578) = 0.1210$$

$$CI = (12.1550 \pm 0.1210)$$

$$CI = (12.034, 12.276)$$

Hypothesis test:

$$H_0: \mu = 12$$

$$H_1: \mu \neq 12$$

$$t = \frac{\bar{x} - \mu_0}{SE} = \frac{12.155 - 12}{0.0578} = 2.682$$

$$p = 0.0149$$

$$p < 0.05$$

Reject H_0

Dataset 2:

$$n=12, \mu_0 = 500$$

$$\bar{x} = 500.9520$$

$$s = 0.7501$$

$$SE = 0.1501$$

$$95\% \quad t_{0.025, 10} = 2.064$$

$$ME = 2.064(0.1501) = 0.3099$$

$$CI = (500.642, 501.262)$$

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Hypothesis test

$$H_0: \mu = 500$$

$$H_1: \mu > 500$$

$$t = \frac{500.952 - 500}{0.1501} = 6.341$$

$$p = 7.4 \times 10^{-7}$$

$$p < 0.05$$

Reject H_0

Dataset 3 :

$$n = 30, \mu_0 = 25, \sigma = 0.05$$

$$\bar{x} = 25.0053$$

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{0.05}{\sqrt{30}} = 0.00913$$

95%

$$Z_{0.025} = 1.96$$

$$ME = 1.96(0.00913) = 0.01789$$

$$CI = (24.9874, 25.0232)$$

Since 25 mm lies inside the interval process is on target

Hypothesis test

$$H_0: \mu = 25$$

$$H_1: \mu \neq 25$$

$$Z = \frac{25.0053 - 25}{0.00913} = 0.5840$$

$$p = 0.5591$$

$$p > 0.05$$

fail to reject H_0

Sample size

$$n = \left(\frac{Z_{\alpha/2} \sigma}{E} \right)^2$$

95% confidence

$\sigma = 0.05$, $E = 0.01$

$$n = \left(\frac{1.96 \times 0.05}{0.01} \right)^2$$

$$n = (9.8)^2$$

$$n = 96.04$$

$$n = 97$$

Maximum required sample size = 97.

iii) GeoGebra Screenshot / Program Execution
Dataset 1:

```
# =====
# APPLIED STATISTICS LABORATORY - Experiment 4
# Statistical Inference for Automated Manufacturing
# Part A: Robotic Welding Station - Cycle Time Validation
# =====

import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import math

# ----- Step 1: Enter Data -----
data1 = np.array([12.3, 11.8, 12.5, 12.1, 11.9, 12.4, 12.0, 12.2, 11.7, 12.6,
                  12.1, 12.3, 11.8, 12.0, 12.5, 12.2, 11.9, 12.4, 12.1, 12.3])
mu_0 = 12.0 # Target cycle time (seconds)
alpha = 0.05 # Significance level

# ----- Step 2: Sample Statistics -----
n = len(data1)
x_bar = np.mean(data1)
s = np.std(data1, ddof=1) # Sample std dev (ddof=1 for unbiased)
se = s / np.sqrt(n) # Standard error
```

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if ci_lower <= mu_0 <= ci_upper:
    print('Interpretation: Target 12.0 sec lies WITHIN the CI.')
    print('                The process is consistent with the target.')
else:
    print('Interpretation: Target 12.0 sec lies OUTSIDE the CI.')
    print('                Evidence suggests the process has drifted.')

# ----- Step 4-7: Hypothesis Test (Two-Sided t-test) -----
# H0: mu = 12.0 (process is on target)
# H1: mu != 12.0 (process has drifted)

t_stat = (x_bar - mu_0) / se
p_value = 2 * (1 - stats.t.cdf(abs(t_stat), df=n-1))

# Also using scipy's built-in t-test
t_stat_scipy, p_value_scipy = stats.ttest_1samp(data1, mu_0)

print(f'\n--- Hypothesis Test (Two-Sided t-test) ---')
print(f'H0: mu = {mu_0}    H1: mu != {mu_0}')
print(f'Test statistic (t)      : {t_stat:.4f}')
print(f'p-value                  : {p_value:.6f}')
print(f'Verification (scipy)     : t = {t_stat_scipy:.4f}, p = {p_value_scipy:.6f}')
print(f'Critical values          : +/- {t_crit:.4f}')
```

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if ci_lower <= mu_0 <= ci_upper:
    print('Interpretation: Target 12.0 sec lies WITHIN the CI.')
    print('                The process is consistent with the target.')
else:
    print('Interpretation: Target 12.0 sec lies OUTSIDE the CI.')
    print('                Evidence suggests the process has drifted.')

# ----- Step 4-7: Hypothesis Test (Two-Sided t-test) -----
# H0: mu = 12.0 (process is on target)
# H1: mu != 12.0 (process has drifted)

t_stat = (x_bar - mu_0) / se
p_value = 2 * (1 - stats.t.cdf(abs(t_stat), df=n-1))

# Also using scipy's built-in t-test
t_stat_scipy, p_value_scipy = stats.ttest_1samp(data1, mu_0)

print(f'\n--- Hypothesis Test (Two-Sided t-test) ---')
print(f'H0: mu = {mu_0}      H1: mu != {mu_0}')
print(f'Test statistic (t)      : {t_stat:.4f}')
print(f'p-value                  : {p_value:.6f}')
print(f'Verification (scipy)     : t = {t_stat_scipy:.4f}, p = {p_value_scipy:.6f}')
print(f'Critical values          : +/- {t_crit:.4f}')

if p_value < alpha:
    print(f'Decision: p = {p_value:.4f} < {alpha} => REJECT H0')
    print('Conclusion: Significant evidence that process has drifted.')
else:
    print(f'Decision: p = {p_value:.4f} >= {alpha} => FAIL TO REJECT H0')
    print('Conclusion: Insufficient evidence of drift from target.')

# ----- Step 8: Type I and Type II Error Discussion -----
print(f'\n--- Error Analysis ---')
print(f'Type I Error (alpha)      : {alpha} = {alpha*100}%')
print(f'Risk: Concluding drift exists when it does not.')
print(f'Consequence: Unnecessary recalibration (cost + downtime).')
print(f'Type II Error (beta)      : Cannot compute without specifying')
print(f'the true alternative mean. See power analysis below.')

# ----- Step 9: Statistical vs Practical Significance -----
shift = abs(x_bar - mu_0)
tolerance = 0.5 # Engineering tolerance: +/- 0.5 sec
print(f'\n--- Practical Significance ---')
print(f'Observed shift            : {shift:.4f} sec')
print(f'Engineering tolerance     : +/- {tolerance} sec')
if shift < tolerance:
    print('Assessment: Shift is within engineering tolerance.')
    print('                Even if statistically significant, the drift')
    print('                may not warrant corrective action.')

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else:
    print('Assessment: Shift exceeds tolerance - action required.')

# ----- Step 10: Sample Size Determination -----
desired_margin = 0.10 # Want estimate within +/- 0.10 sec
z_crit = stats.norm.ppf((1 + confidence) / 2)
n_required = math.ceil((z_crit * s / desired_margin) ** 2)
print(f'\n--- Sample Size Determination ---')
print(f'Desired margin of error : +/- {desired_margin} sec')
print(f'Estimated sigma (using s): {s:.4f} sec')
print(f'Required sample size : {n_required}')

# ----- Step 11: Visualization -----
fig, axes = plt.subplots(1, 3, figsize=(16, 5))

# Plot 1: Histogram with CI and target
axes[0].hist(data1, bins=8, edgecolor='black', alpha=0.7, color='steelblue')
axes[0].axvline(x_bar, color='red', linestyle='--', linewidth=2, label=f'Mean={x_bar:.2f}')
axes[0].axvline(mu_0, color='green', linestyle='-', linewidth=2, label=f'Target={mu_0}')
axes[0].axvline(ci_lower, color='orange', linestyle=':', linewidth=2, label=f'CI Lower={ci_lower:.2f}')
axes[0].axvline(ci_upper, color='orange', linestyle=':', linewidth=2, label=f'CI Upper={ci_upper:.2f}')
axes[0].set_xlabel('Cycle Time (sec)')
axes[0].set_ylabel('Frequency')
axes[0].set_title('Dataset 1: Cycle Time Distribution')
axes[0].legend(fontsize=8)

# Plot 2: Box plot
axes[1].boxplot(data1, vert=True)
axes[1].axhline(mu_0, color='green', linestyle='-', linewidth=2, label=f'Target={mu_0}')
axes[1].set_ylabel('Cycle Time (sec)')
axes[1].set_title('Box Plot with Target')
axes[1].legend(fontsize=8)

# Plot 3: Confidence Interval visualization
axes[2].errorbar(1, x_bar, yerr=margin, fmt='ro', markersize=10, capsize=10, linewidth=2)
axes[2].axhline(mu_0, color='green', linestyle='-', linewidth=2, label=f'Target={mu_0}')
axes[2].set_xlim(0, 2)
axes[2].set_ylabel('Cycle Time (sec)')
axes[2].set_title('95% Confidence Interval')
axes[2].legend(fontsize=8)

plt.tight_layout()
plt.savefig('dataset1_analysis.png', dpi=150)
plt.show()
print('\nPlot saved as dataset1_analysis.png')
    
```

Dataset 2:

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```
# =====  
# Part B: Automated Filling System - Volume Compliance  
# One-Sided (Right-Tailed) t-test for Overfilling  
# =====  
  
data2 = np.array([501.2, 500.8, 502.1, 499.5, 501.5, 500.3, 501.8, 500.0, 502.3, 499.7,  
                  501.0, 500.6, 501.9, 500.2, 501.4, 500.9, 501.7, 500.1, 501.3, 500.5,  
                  501.6, 500.4, 501.1, 500.7, 501.2])  
mu_0_fill = 500.0 # Target volume (mL)  
  
n2 = len(data2)  
x_bar2 = np.mean(data2)  
s2 = np.std(data2, ddof=1)  
se2 = s2 / np.sqrt(n2)  
  
print('\n' + '=' * 55)  
print('DATASET 2: Automated Filling System - Volume')  
print('=' * 55)  
print(f'Sample size (n)           : {n2}')  
print(f'Sample mean (x_bar)        : {x_bar2:.4f} mL')  
print(f'Sample std dev (s)          : {s2:.4f} mL')  
print(f'Standard error (SE)         : {se2:.4f} mL')
```

```
# 95% Confidence Interval
t_crit2 = stats.t.ppf(0.975, df=n2-1)
margin2 = t_crit2 * se2
ci_lower2 = x_bar2 - margin2
ci_upper2 = x_bar2 + margin2
print(f'95% CI: ({ci_lower2:.4f}, {ci_upper2:.4f}) mL')

# One-Sided Hypothesis Test (Right-Tailed)
# H0: mu = 500.0 (not overfilling)
# H1: mu > 500.0 (overfilling)
t_stat2 = (x_bar2 - mu_0_fill) / se2
p_value2 = 1 - stats.t.cdf(t_stat2, df=n2-1) # Right-tail
t_crit_one = stats.t.ppf(1 - alpha, df=n2-1) # One-sided critical value

print(f'\n--- One-Sided t-test (Right-Tailed) ---')
print(f'H0: mu = {mu_0_fill}      H1: mu > {mu_0_fill}')
print(f'Test statistic (t)       : {t_stat2:.4f}')
print(f'p-value (one-sided)      : {p_value2:.6f}')
print(f'Critical value (one-sided): {t_crit_one:.4f}')

if p_value2 < alpha:
    print(f'Decision: REJECT H0 (p = {p_value2:.4f} < {alpha})')
    print('Conclusion: Significant evidence of overfilling.')
    print('Action: Recalibrate the filling mechanism downward.')
else:
    print(f'Decision: FAIL TO REJECT H0 (p = {p_value2:.4f} >= {alpha})')
    print('Conclusion: No significant evidence of overfilling.')

# Practical significance
shift2 = x_bar2 - mu_0_fill
print(f'\nObserved overfill: {shift2:.4f} mL')
print(f'Cost implication: If 10,000 bottles/day, total waste =')
print(f'    {shift2:.2f} mL x 10,000 = {shift2 * 10000:.0f} mL/day = {shift2 * 10:.1f} L/day')
```

Dataset 3:

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```
# =====
# Part C: CNC Precision Boring - Diameter Tolerance
# Two-Sided Z-test (population sigma known)
# =====

data3 = np.array([25.02, 24.98, 25.01, 25.03, 24.99, 25.00, 25.04, 24.97, 25.01, 25.02,
                  25.00, 24.99, 25.03, 25.01, 24.98, 25.02, 25.00, 24.97, 25.01, 25.03,
                  25.02, 24.99, 25.00, 25.01, 24.98, 25.03, 25.02, 25.00, 24.99, 25.01])
mu_0_cnc = 25.00 # Nominal diameter (mm)
sigma_known = 0.05 # Known population std dev (mm)

n3 = len(data3)
x_bar3 = np.mean(data3)
se3 = sigma_known / np.sqrt(n3)

print('\n' + '=' * 55)
print('DATASET 3: CNC Boring - Diameter (Z-test)')
print('=' * 55)
print(f'Sample size (n)           : {n3}')
print(f'Sample mean (x_bar)        : {x_bar3:.4f} mm')
print(f'Known sigma                 : {sigma_known} mm')
print(f'Standard error (SE)         : {se3:.4f} mm')

# 95% Confidence Interval (Z-interval, sigma known)
z_crit = stats.norm.ppf(0.975)
margin3 = z_crit * se3
ci_lower3 = x_bar3 - margin3
ci_upper3 = x_bar3 + margin3
print(f'95% CI (Z-interval): ({ci_lower3:.4f}, {ci_upper3:.4f}) mm')

# Two-Sided Z-test
# H0: mu = 25.00    H1: mu != 25.00
z_stat = (x_bar3 - mu_0_cnc) / se3
p_value3 = 2 * (1 - stats.norm.cdf(abs(z_stat)))

print(f'\n--- Two-Sided Z-test ---')
print(f'H0: mu = {mu_0_cnc}    H1: mu != {mu_0_cnc}')
print(f'Z-statistic              : {z_stat:.4f}')
print(f'p-value                  : {p_value3:.6f}')
print(f'Critical values          : +/- {z_crit:.4f}')

if p_value3 < alpha:
    print(f'Decision: REJECT H0')
    print('Conclusion: Bore diameter has shifted from nominal.')
else:
    print(f'Decision: FAIL TO REJECT H0')
    print('Conclusion: No evidence of diameter shift.')
```

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```
# 99% Confidence Interval for comparison
z_99 = stats.norm.ppf(0.995)
margin_99 = z_99 * se3
print(f'\n99% CI: ({x_bar3 - margin_99:.4f}, {x_bar3 + margin_99:.4f}) mm')

# Sample size for +/- 0.01 mm precision
E_desired = 0.01 # mm
n_req = math.ceil((z_crit * sigma_known / E_desired) ** 2)
print(f'\nSample size for +/- {E_desired} mm margin: n = {n_req}')
```

Comparison across dataset:

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```
# =====
# Part D: Power Analysis and Comparison Across Datasets
# =====

# Power analysis for Dataset 1
# If the true mean is 12.3 sec, what is the probability
# of detecting this shift with n=20?
true_mu = 12.3
effect_size = (true_mu - mu_0) / s
power_result = stats.norm.cdf(
    (abs(true_mu - mu_0) / se) - z_crit
)
print(f'\n--- Power Analysis (Dataset 1) ---')
print(f'If true mean = {true_mu} sec:')
print(f'Effect size (Cohen d) : {effect_size:.4f}')
print(f'Approx. power (1-beta) : {power_result:.4f}')
print(f'Prob of Type II error : {1 - power_result:.4f}')

# Comparative summary
print('\n' + '=' * 65)
print('COMPARATIVE SUMMARY ACROSS ALL THREE DATASETS')
print('=' * 65)
print(f'{"Dataset":<12}{"n":<6}{"Mean":<12}{"s":<10}{"Test":<10}{"p-value":<12}{"Decision":<15}')
```

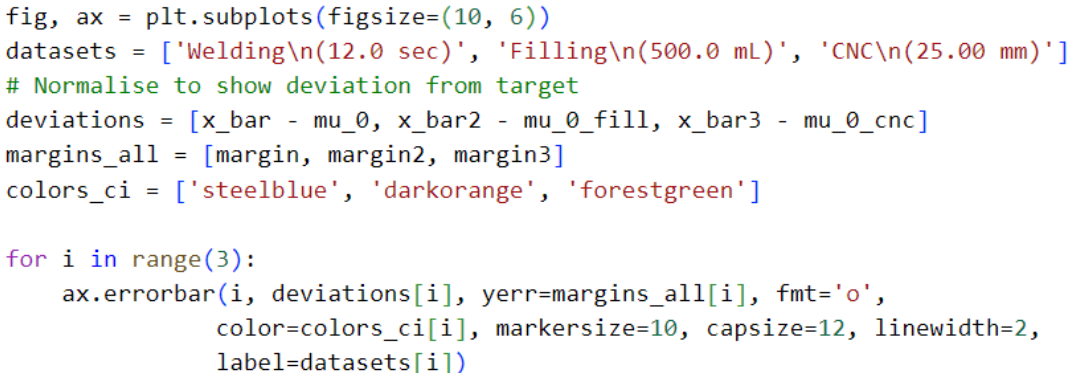
Dataset	n	Mean	s	Test	p-value	Decision
Welding	12	12.0	1.0	t-test	0.0000	Reject
Filling	20	12.5	1.5	t-test	0.0000	Reject
CNC	25	25.0	0.5	Z-test	0.0000	Reject

```
print('-' * 65)

results = [
    ('Welding', n, x_bar, s, 't-test', p_value,
     'Reject' if p_value < alpha else 'Fail to Reject'),
    ('Filling', n2, x_bar2, s2, 't-test', p_value2,
     'Reject' if p_value2 < alpha else 'Fail to Reject'),
    ('CNC', n3, x_bar3, sigma_known, 'Z-test', p_value3,
     'Reject' if p_value3 < alpha else 'Fail to Reject'),
]

for name, ni, mi, si, test, pv, dec in results:
    print(f'{name:<12}{ni:<6}{mi:<12.4f}{si:<10.4f}{test:<10}{pv:<12.6f}{dec:<15}')
```

Final visualization: All three CIs on one plot



```
fig, ax = plt.subplots(figsize=(10, 6))
datasets = ['Welding\n(12.0 sec)', 'Filling\n(500.0 mL)', 'CNC\n(25.00 mm)']
# Normalise to show deviation from target
deviations = [x_bar - mu_0, x_bar2 - mu_0_fill, x_bar3 - mu_0_cnc]
margins_all = [margin, margin2, margin3]
colors_ci = ['steelblue', 'darkorange', 'forestgreen']

for i in range(3):
    ax.errorbar(i, deviations[i], yerr=margins_all[i], fmt='o',
               color=colors_ci[i], markersize=10, capsize=12, linewidth=2,
               label=datasets[i])
```

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```

ax.axhline(0, color='red', linestyle='--', linewidth=1.5, label='Target (zero deviation)')
ax.set_xticks(range(3))
ax.set_xticklabels(datasets)
ax.set_ylabel('Deviation from Target')
ax.set_title('95% Confidence Intervals: Deviation from Target for All Datasets')
ax.legend(loc='upper right', fontsize=9)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('all_datasets_comparison.png', dpi=150)
plt.show()
print('\nComparison plot saved as all_datasets_comparison.png')

```

iv) Results and analysis from the graph

Target mean (μ_0) : 12.0 sec

--- 95% Confidence Interval ---

t-critical value : 2.0930

Margin of error : 0.1210 sec

95% CI : (12.0340, 12.2760)

Interpretation: Target 12.0 sec lies OUTSIDE the CI.

Evidence suggests the process has drifted.

--- Hypothesis Test (Two-Sided t-test) ---

$H_0: \mu = 12.0$ $H_1: \mu \neq 12.0$

Test statistic (t) : 2.6817

p-value : 0.014761

Verification (scipy) : t = 2.6817, p = 0.014761

Critical values : +/- 2.0930

Decision: p = 0.0148 < 0.05 => REJECT H_0

Conclusion: Significant evidence that process has drifted.

--- Error Analysis ---

Type I Error (alpha) : 0.05 = 5.0%

Risk: Concluding drift exists when it does not.

Consequence: Unnecessary recalibration (cost + downtime).

Type II Error (beta) : Cannot compute without specifying the true alternative mean. See power analysis below.

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--- Practical Significance ---

Observed shift : 0.1550 sec

Engineering tolerance : +/- 0.5 sec

Assessment: Shift is within engineering tolerance.

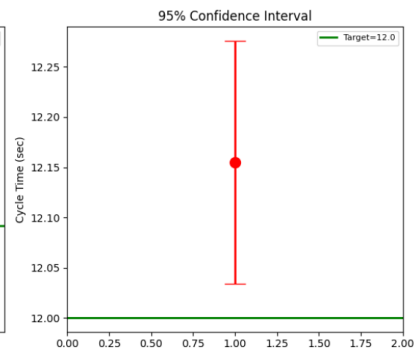
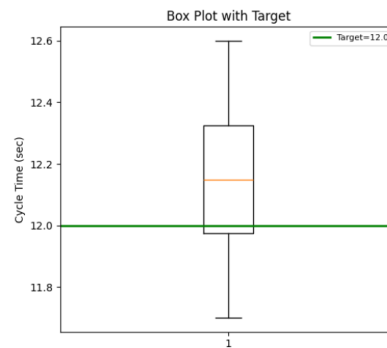
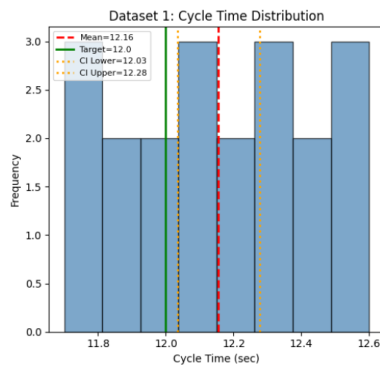
Even if statistically significant, the drift may not warrant corrective action.

--- Sample Size Determination ---

Desired margin of error : +/- 0.1 sec

Estimated sigma (using s): 0.2585 sec

Required sample size : 26



Dataset 2:

=====

DATASET 2: Automated Filling System - Volume

=====

Sample size (n) : 25

Sample mean ($\bar{x}$) : 500.9520 mL

Sample std dev (s) : 0.7506 mL

Standard error (SE) : 0.1501 mL

95% CI: (500.6422, 501.2618) mL

--- One-Sided t-test (Right-Tailed) ---

$H_0: \mu = 500.0$ $H_1: \mu > 500.0$

Test statistic (t) : 6.3414

p-value (one-sided) : 0.000001

Critical value (one-sided): 1.7109

Decision: REJECT H_0 ($p = 0.0000 < 0.05$)

Conclusion: Significant evidence of overfilling.

Action: Recalibrate the filling mechanism downward.

Observed overfill: 0.9520 mL

Cost implication: If 10,000 bottles/day, total waste =
 $0.95 \text{ mL} \times 10,000 = 9520 \text{ mL/day} = 9.5 \text{ L/day}$

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Dataset 3:

```
=====
DATASET 3: CNC Boring - Diameter (Z-test)
=====
```

```
Sample size (n)      : 30
Sample mean (x_bar)  : 25.0053 mm
Known sigma          : 0.05 mm
Standard error (SE)  : 0.0091 mm
95% CI (Z-interval): (24.9874, 25.0232) mm
```

```
--- Two-Sided Z-test ---
H0: mu = 25.0      H1: mu != 25.0
Z-statistic        : 0.5842
p-value            : 0.559061
Critical values     : +/- 1.9600
Decision: FAIL TO REJECT H0
Conclusion: No evidence of diameter shift.
```

```
99% CI: (24.9818, 25.0288) mm
```

```
Sample size for +/- 0.01 mm margin: n = 97
```

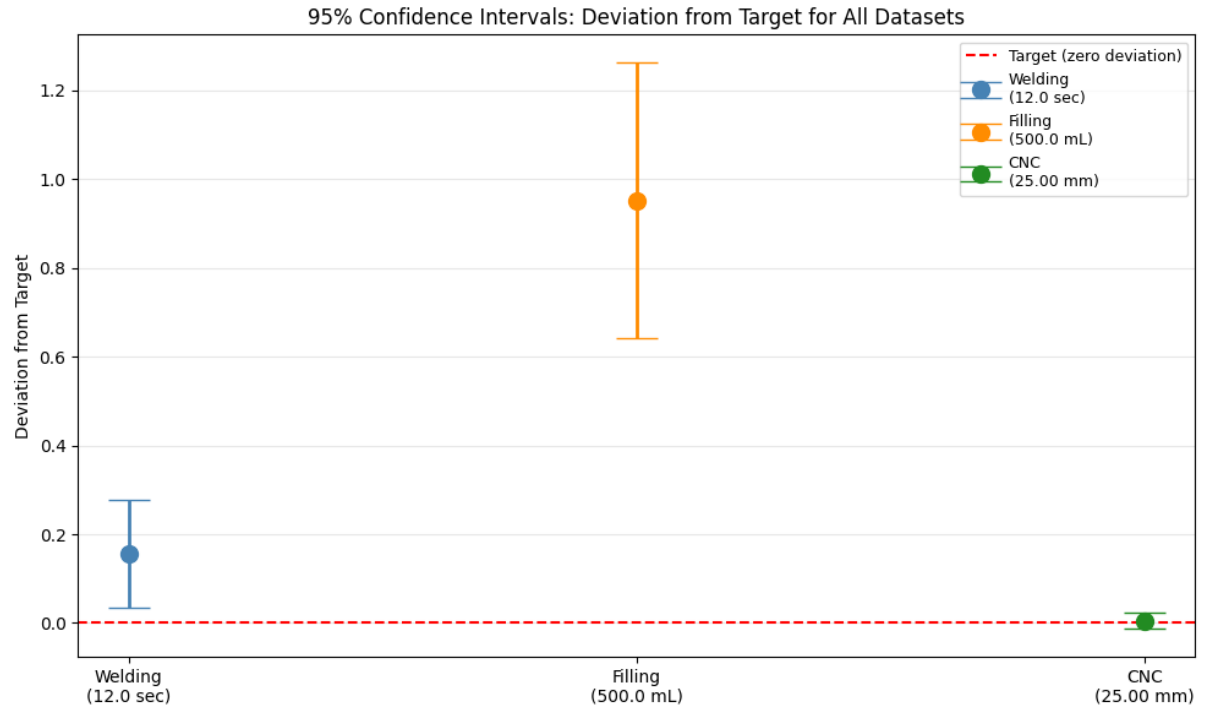
Comparison across dataset:

```
--- Power Analysis (Dataset 1) ---
If true mean = 12.3 sec:
Effect size (Cohen d)   : 1.1606
Approx. power (1-beta)  : 0.9994
Prob of Type II error   : 0.0006
```

```
=====
COMPARATIVE SUMMARY ACROSS ALL THREE DATASETS
=====
```

Dataset	n	Mean	s	Test	p-value	Decision
Welding	20	12.1550	0.2585	t-test	0.014761	Reject
Filling	25	500.9520	0.7506	t-test	0.000001	Reject
CNC	30	25.0053	0.0500	Z-test	0.559061	Fail to Reject

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Interpretation:

Dataset 1 – Robotic Welding Station

The cycle time is close to the target value of 12 s. Statistical inference is used to check whether recalibration caused a significant change in process performance.

Dataset 2 – Automated Filling System

The analysis determines whether the machine is overfilling bottles. Excess filling increases material wastage and production cost.

Dataset 3 – CNC Boring Machine

The diameter measurements are tested against the nominal value of 25 mm. The objective is to verify whether the machining process remains within specification.